pioneer Konstantin Tsiolkovsky first envisioned, we could employ solar sails—vast but very thin films that catch sunlight and the solar wind, a caravel kilometers wide plying the void between the worlds. Especially for trips to Mars and beyond, such methods are far better than rockets.

As with most technologies, when something barely works, when it's the first of its kind, there's a natural tendency to improve it, develop it, exploit it. Soon there's such an institutional investment in the original technology, no matter how flawed, that it's very hard to move on to something better. NASA has almost no resources to pursue alternative propulsion technologies. That money would have to come out of near-term missions, missions which could provide concrete results and improve NASA's success record. Spending money on alternative technologies pays off a decade or two in the future. We tend to be very little interested in a decade or two in the future. This is one of the ways by which initial success can sow the seeds of ultimate failure; and is very similar to what sometimes happens in biological evolution. But sooner or later some nation—perhaps one without a huge investment in marginally effective technology—will develop effective alternatives.

Even before then, if we take a cooperative path, there will come a time—perhaps in the first decades of the new century and the new millennium—when an interplanetary spacecraft is assembled in Earth orbit, the progress in full view on the evening news. Astronauts and cosmonauts, hovering like gnats, guide and mate the prefabricated parts. Eventually the ship, tested and ready, is boarded by its international crew, and boosted to escape velocity. For the whole of the voyage to Mars and back, the lives of the crew members depend on one another, a microcosm of our actual circumstances down here on Earth. Perhaps the first joint interplanetary mission with human crews will be only a flyby or orbit of Mars. Earlier, robot vehicles, with aerobraking, parachutes, and retrorockets, will have set gently down on the Martian surface to collect samples and return them to Earth, and to emplace supplies for future explorers. But whether or not we have compelling, coherent reasons, I am sure—unless we destroy ourselves first—that